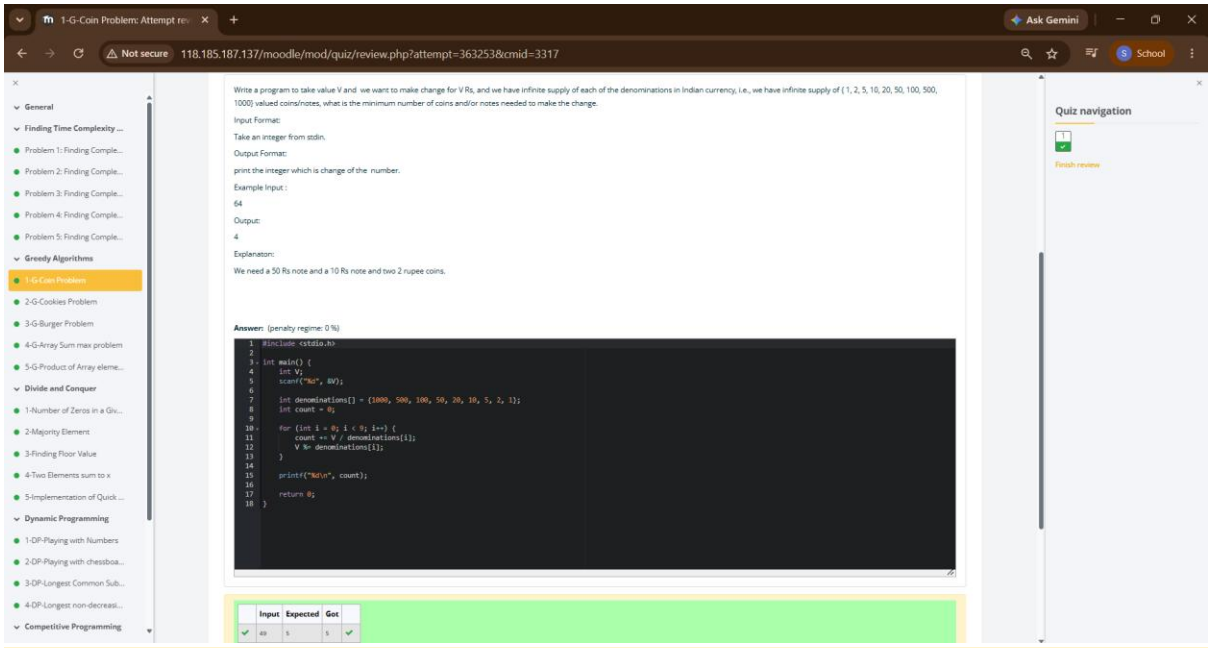
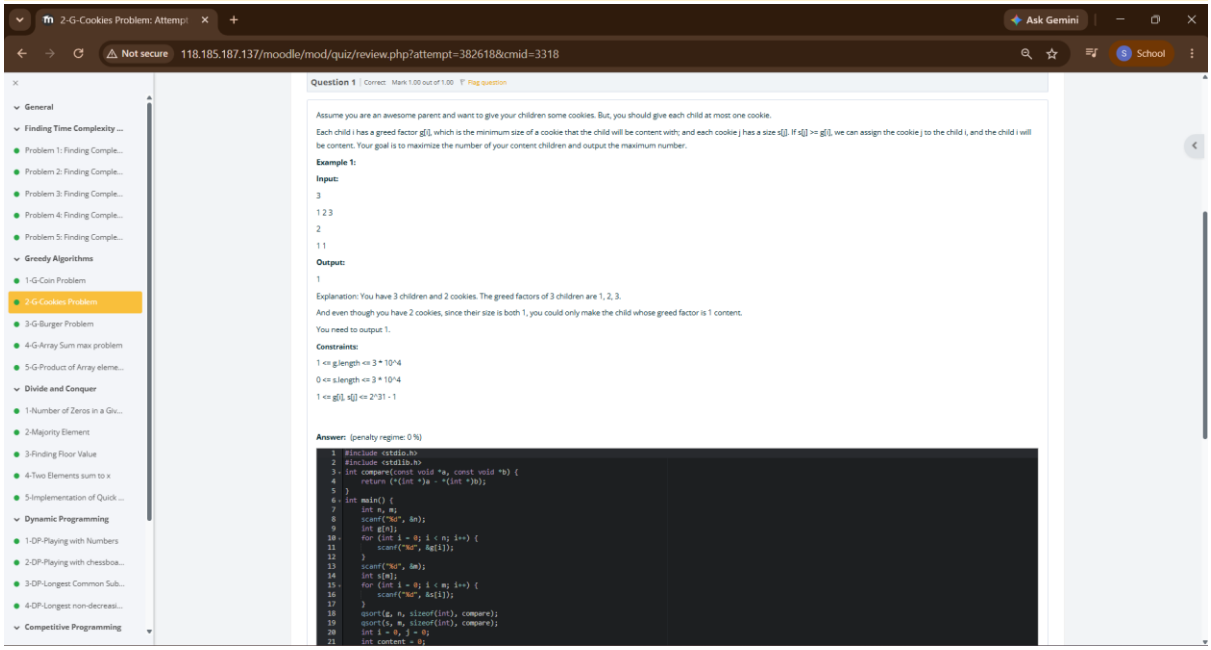
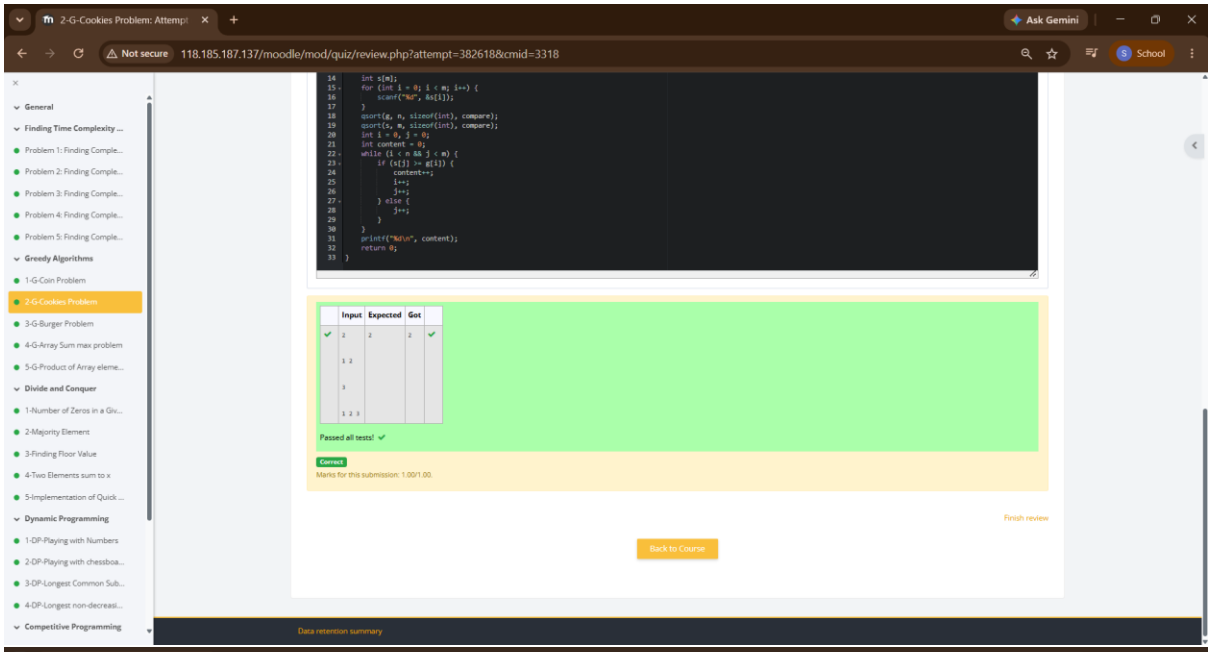
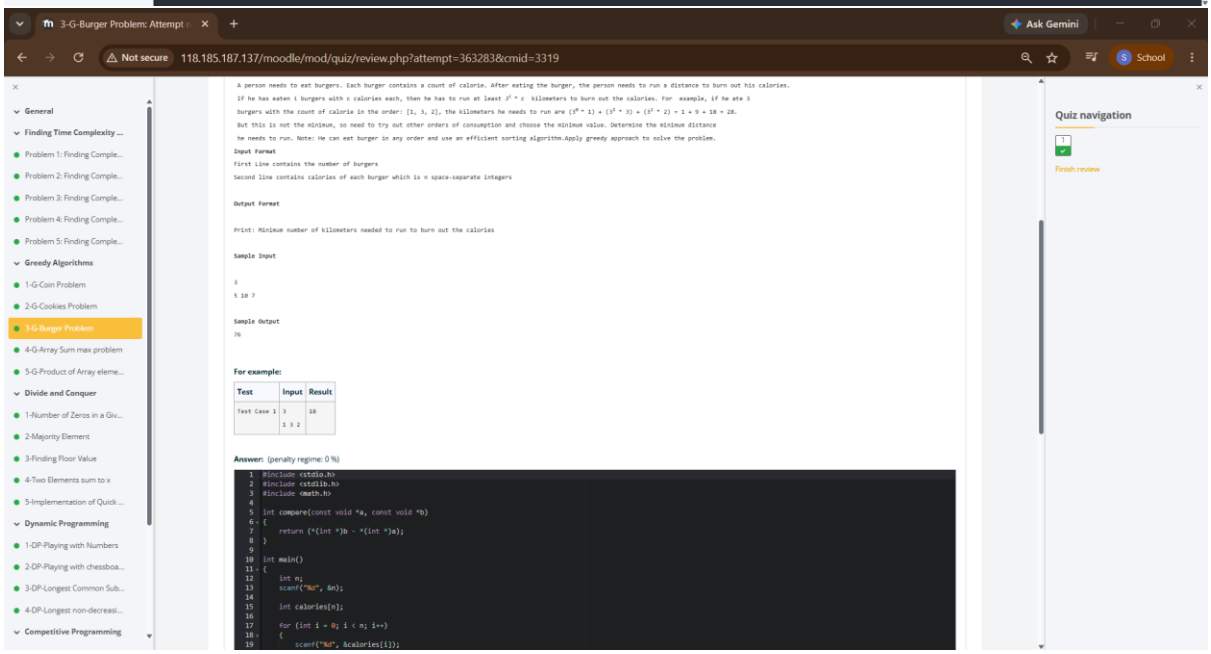
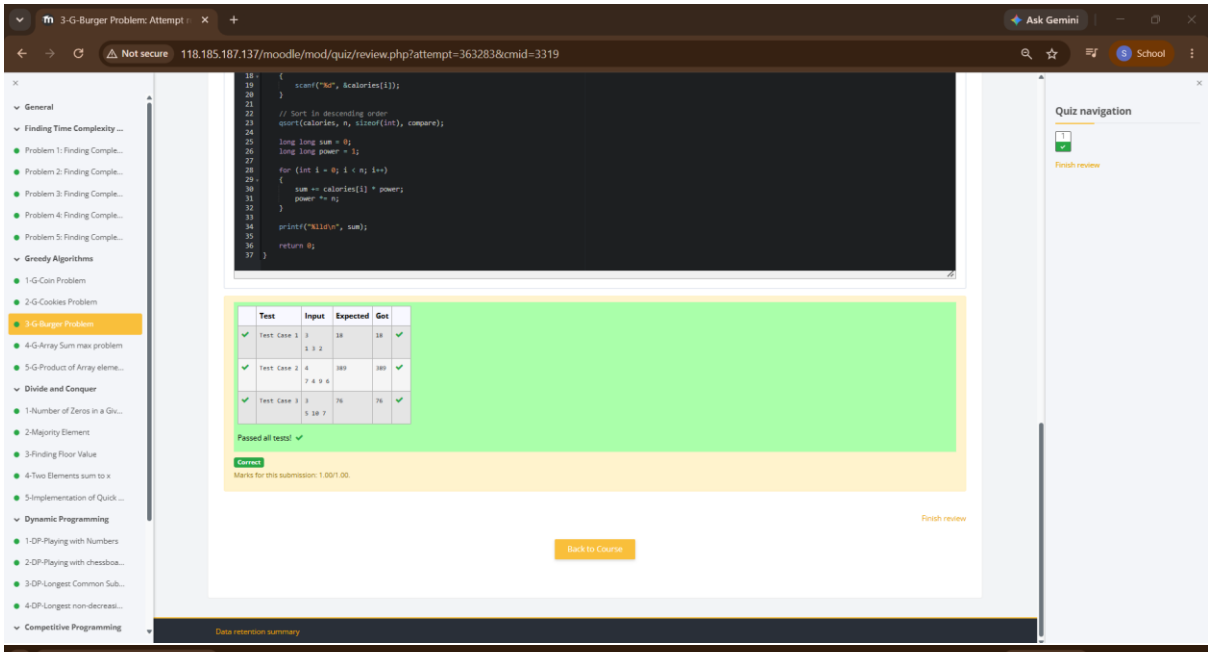
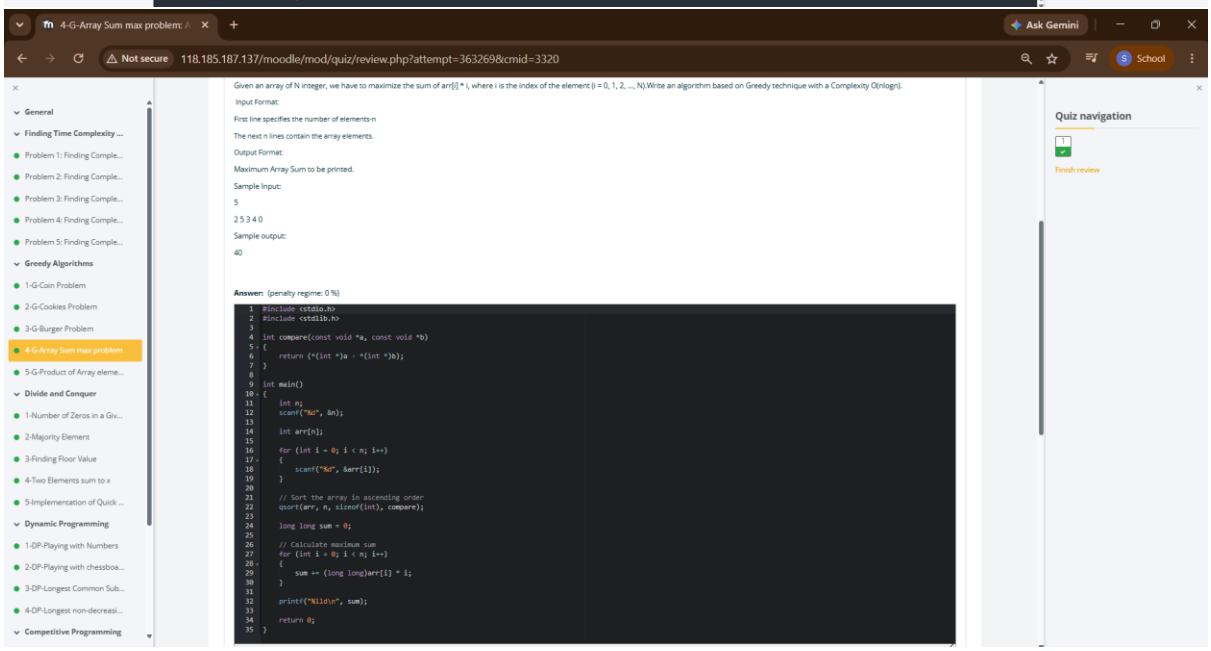

1-G-Coin Problem: Attempt rev
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=363253&cmid=3317
Ask Gemini
School
Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.
Input Format:
Take an integer from stdin.
Output Format:
print the integer which is change of the number.
Example Input :
64
Output:
4
Explanation:
We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.
Answer: (penalty regime: 0 %)
#include <stdio.h>
int main() {
int V;
scanf("%d", &V);
int denominations[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
int count = 0;
for (int i = 0; i < 9; i++) {
count += V / denominations[i];
V %= denominations[i];
}
printf("%d\n", count);
return 0;
}
Quiz navigation
Finish review
Input
Expected
Got
49
5
5
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.


2-G-Cookies Problem: Attempt
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=382618&cmid=3318
Ask Gemini
School
Question 1
Correct
Mark 1.00 out of 1.00
Flag question
Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.
Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.
Example 1:
Input:
3
1 2 3
2
1 1
Output:
1
Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.
And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.
You need to output 1.
Constraints:
1 <= g.length <= 3 * 10^4
0 <= s.length <= 3 * 10^4
1 <= g[i], s[j] <= 2^31 - 1
Answer: (penalty regime: 0 %)
#include <stdio.h>
#include <stdlib.h>
int compare(const void *a, const void *b) {
return (*(int *)a - *(int *)b);
}
int main() {
int n, m;
scanf("%d", &n);
int g[n];
for (int i = 0; i < n; i++) {
scanf("%d", &g[i]);
}
scanf("%d", &m);
int s[m];
for (int i = 0; i < m; i++) {
scanf("%d", &s[i]);
}
qsort(g, n, sizeof(int), compare);
qsort(s, m, sizeof(int), compare);
int i = 0, j = 0;
int content = 0;


2-G-Cookies Problem: Attempt
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=382618&cmid=3318
Ask Gemini
School
General
Finding Time Complexity ...
Problem 1: Finding Comple...
Problem 2: Finding Comple...
Problem 3: Finding Comple...
Problem 4: Finding Comple...
Problem 5: Finding Comple...
Greedy Algorithms
1-G-Coin Problem
2-G-Cookies Problem
3-G-Burger Problem
4-G-Array Sum max problem
5-G-Product of Array eleme...
Divide and Conquer
1-Number of Zeros in a Giv...
2-Majority Element
3-Finding Floor Value
4-Two Elements sum to x
5-Implementation of Quick ...
Dynamic Programming
1-DP-Playing with Numbers
2-DP-Playing with chessboa...
3-DP-Longest Common Sub...
4-DP-Longest non-decreasi...
Competitive Programming
int s[m];
for (int i = 0; i < m; i++) {
scanf("%d", &s[i]);
}
qsort(g, n, sizeof(int), compare);
qsort(s, m, sizeof(int), compare);
int i = 0, j = 0;
int content = 0;
while (i < n && j < m) {
if (s[j] >= g[i]) {
content++;
i++;
j++;
} else {
j++;
}
}
printf("%d\n", content);
return 0;
}
Input
Expected
Got
2
2
2
1 2
3
1 2 3
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.
Finish review
Back to Course
Data retention summary


3-G-Burger Problem: Attempt
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=363283&cmid=3319
Ask Gemini
School
A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.
If he has eaten t burgers with c calories each, then he has to run at least 3^t * c kilometers to burn out the calories. For example, if he ate 3
burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are (3^1 * 1) + (3^3 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28.
But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance
he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.
Input Format
First line contains the number of burgers
Second line contains calories of each burger which is n space-separate integers
Output Format
Print: Minimum number of kilometers needed to run to burn out the calories
Sample Input
3
5 10 7
Sample Output
76
For example:
Test
Input
Result
Test Case 1
3
1 3 2
18
Answer: (penalty regime: 0 %)
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
int compare(const void *a, const void *b)
{
return (*(int *)b - *(int *)a);
}
int main()
{
int n;
scanf("%d", &n);
int calories[n];
for (int i = 0; i < n; i++)
{
scanf("%d", &calories[i]);
Quiz navigation
Finish review


3-G-Burger Problem: Attempt
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=363283&cmid=3319
Ask Gemini
School
Quiz navigation
Finish review
Test
Input
Expected
Got
Test Case 1
Test Case 2
Test Case 3
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.
Finish review
Back to Course
Data retention summary

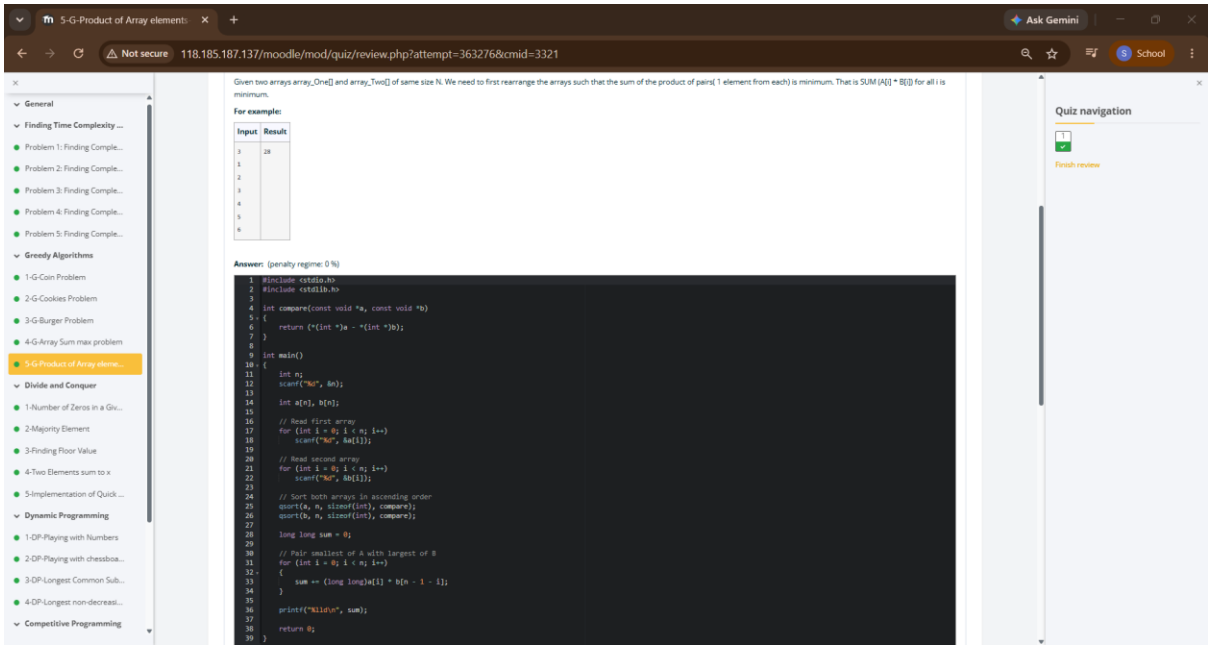

4-G-Array Sum max problem:
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=363269&cmid=3320
Ask Gemini
School
Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).
Input Format:
First line specifies the number of elements-n
The next n lines contain the array elements.
Output Format:
Maximum Array Sum to be printed.
Sample Input:
5
2 5 3 4 0
Sample output:
40
Answer: (penalty regime: 0 %)
Quiz navigation
Finish review

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.



	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.